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IOT Smart Car Parking System

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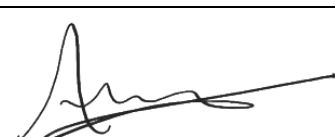
Declaration

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Project Details

Project Title	IOT Smart Car Parking System
Project ID	PEP_13

Group Members

Reg. No	Name	Signature
IT21293344	Wickramaarachchi K. A. S. N.	
IT21300196	Manathunga M. D. T. L.	
IT21325700	Bandara H. M. S. I. D.	
IT21293108	Bogahawatta C. A.	

Abstract

Parking has become a significant problem in metropolitan areas due to expanding urbanization and the increased number of vehicles on the road. Traditional parking systems frequently experience inefficiencies, a lack of available parking spaces, and trouble locating open spaces. We suggest a novel approach to address these issues that enhance the parking experience using Internet of Things (IoT) technologies. The system is easy to use and navigate, giving drivers a hassle-free parking experience. The system precisely determines the parking space availability by using sensors to identify the presence of cars in parking spaces. The information is shown on a user interface in their smartphone for drivers to access. The system also uses RFID cards for authentication. It additionally protects the system. The effectiveness of parking was significantly increased due to the deployment, which also reduced traffic congestion brought on by drivers looking for parking and increased customer convenience. Additionally, the system supplied helpful information about parking patterns that parking managers and urban planners can use to plan and make decisions.

Acknowledgement

Project success needs a lot of direction and help from numerous people. First and foremost, we would like to express our sincere gratitude to Ms. Narmada Gamage, our lecturer, for providing us with the beautiful chance to complete this project. We appreciate that we could do this project in the allotted amount of time from the lecture. With the effort and cooperation of our group members, this project can be finished. They did their best to assist with the project's completion by offering much encouragement to maintain this pace. Furthermore, we are humbled to recognize our depth through my friends, who helped me in purchasing high-quality equipment.

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List of Acronyms and Abbreviations

RFID	Radio Frequency Identification
NodeMCU	Node Microcontroller Unit
Arduino IDE	Arduino Integrated Development Environment
ESP8266	Arduino Wi-Fi Module

1. Introduction

1.1 Problem Statement

Metropolitan regions have many obstacles when it comes to parking, such as inefficiency, a lack of parking places, and difficulties finding open spaces. An IoT-based intelligent automobile parking system is suggested to solve these problems. This project will use IoT technology to deliver real-time parking availability information and maximize parking space utilization. Traditional parking systems need more real-time monitoring and efficient allocation of parking spaces, leading to increased congestion and frustration for drivers. The existing parking infrastructure needs to be improved to accommodate the growing number of vehicles, resulting in a shortage of parking spaces. Also, there are so many security issues in Traditional parking systems. The system also uses RFID cards for authentication, and it safeguards the system. Due to a lack of reliable and current information, drivers also need help locating open parking spaces. The Internet of Things (IoT)-based intelligent auto parking system aims to address these issues and enhance the urban parking experience.

1.2 Product Scope

1.2.1 In-scope

1.2.1.1 Monitoring Slot Availability

We provide a LCD at the entry to monitor the open lots and determine whether to enter the parking lot.

1.2.1.2 Display slots on the Blynk application in real-time

We provide a Blynk application to monitor the open lots and determine whether to enter the parking lot.

1.2.1.3 Auto Gate Opening System

The mechanism is set up so that the exit gate will open after payment has been made, and the entrance gate will only open if there are free parking spaces.

1.2.1.4 RFID Authentication

The system includes an RFID authentication system to verify and authorize vehicles entering and exiting the parking facility.

1.2.1.5 Night Lighting System

The product features a fully automated night lighting system for the parking area to ensure visibility and security during the nighttime.

1.2.2 Out-scope

1.2.2.1 Construction of Parking Facility

The project does not include the construction or physical set-up of the parking facility itself. It assumes the availability of a suitable parking area.

1.2.2.2 Vehicle Parking Guidance System

The product does not provide a vehicle parking guidance system like automated parking space allocation or guiding vehicles to specific parking spots.

1.2.2.3 Payment Integration

While the automated fee calculation system is included, the product does not integrate with external payment systems or handle actual financial transactions. It focuses on displaying the fee to the user.

1.2.2.4 Security Surveillance

The project does not cover extensive security surveillance systems like CCTV cameras, advanced security monitoring, or automated license plate recognition.

1.2.2.5 Maintenance and Support

The ongoing maintenance and support of the product, including hardware repairs or system upgrades, are considered out of scope.

1.3 Project Report Structure

Five primary chapters—Introduction, Methodology, Evaluation, Conclusion, and References—make up this project report. Chapter 01 describes the problem that led to this project. It outlines a thorough project specification. The project scope is broken down into in-scope and out-of-scope components in the product scope. The two types of scope define the aspects of the project on which we will and will not focus. Chapter 02, Methodology, has subparts such as requirements and Analysis, design, implementation, and testing. Requirements and Analysis detail the procedures followed to create this project and how we obtained the project's needs. Functional and non-functional requirements and use case diagrams that describe our project are also mentioned here. The description section includes a brief description of the charts used to describe our product, such as activity diagrams, class diagrams, interface diagrams, and circuit diagrams. Under the implementation topic, the physical circuit diagram, the steps that we followed to build the product, the development of the project, modules and functions created, validations, and unique code segments of the circuit is described. Testing describes the test plan and provides evidence that all aspects of the system have been tested. The project work's evaluation can be found in Chapter 03, Evaluation. Whether we successfully get all the anticipated results will determine this. Considering the complexity of specific suggested solutions, we should also consider the implementation lessons we have acquired. This chapter also includes the future implementations we hope to offer clients. Chapter 4, Conclusion, describes how the client got the solution from the project, compared the problems, and defined whether they achieved the target. The limitations of the proposed technique, solutions for them, and the benefits of developing this project for the client are also contained here. The references used for this project report are listed in Chapter 05.

2. Methodology

2.1 Requirements and Analysis

2.1.1 Functional Requirements

- ✓ Functional Requirement 1 - Slot Monitoring System Implementation

Table 1 : Functional Requirement 1

Input	Monitor free slots using IR sensors
Process	Provide Display with free parking spots
Output	Displaying them on an LCD display

- ✓ Functional Requirement 2 – Display slots on the Blynk application in real-time

Table 2 : Functional Requirement 2

Input	Get the result from IR sensors
Process	Provide IR result for Blynk server via ESP8266 module
Output	Displaying them on Blynk application interface

- ✓ Functional Requirement 3 - Automatic Gate Opening System Implementation

Table 3 : Functional Requirement 3

Input	Detecting an approaching car at the gate
Process	Check if there are any parking slots free
Output	If there are just available slots, open the gate.

✓ Functional Requirement 4 - Automatic Night Lighting System Implementation

Table 4 : Functional Requirement 4

Input	Sensing whether it is dark by the LDR sensor
Process	Check if it is dark
Output	Turn on the lighting system in the parking

✓ Functional Requirement 5 - RFID Authentication System Implementation

Table 5 : Functional Requirement 5

Input	Scan the RFID cards
Process	Check if it valid to access
Output	Open the gates and user can enter the parking

2.1.2 Non-Functional Requirements

Performance

- The system must update the parking lot available on the GUI with minimal latency.
- The fee calculation process should be fast and efficient to minimize user waiting time at the exit point.
- The night lighting system should swiftly accommodate changes in ambient light levels.
- Wi-Fi service is necessary for the system to communicate with a mobile device.

Security

- The system must guarantee the security of user data, such as car details and payment information.
- The RFID authentication system shall prevent unauthorized access to the parking facility.
- The system must be protected with the proper safeguards against potential online threats and hacking attempts.

Scalability

- The system should be scalable to support many parking lots and handle growing user traffic.
- It should support future expansion and the addition of new features or functionalities.

2.1.3 Use case scenario

Table 6 : Use Case Scenario 1

Number	01	
Name	Slot Monitoring System Implementation	
Summary	The method of detecting cars by implementing IR sensors and an Atmega328p Arduino board.	
Precondition	Before the coding, connect the IR sensor and the Atmega328p Arduino board.	
Postcondition	Sending a signal to the LCD.	
Primary actor	IR sensors, ATmega328p Arduino board.	
Trigger	Decide is the signal coming from the LCD or not.	
Main Scenario	Step 1	Connect the IR sensors to the Arduino board.
	Step 2	Code the Arduino to detect cars using C programming language.
	Step 3	Set a particular point of sensitivity to detect cars.
	Step 4	Code the Arduino to send a signal to the LCD.
Extension	Step 2.1	Fail to detect cars.
	Step 4.1	Sending a signal to the LCD may fail.

Table 7 : Use Case Scenario 2

Number	02	
Name	Display slots on the Blynk application in real-time	
Summary	The method of detecting slots by implementing IR sensors, an Atmega328p Arduino board and an ESP8266 module.	
Precondition	Before the coding, connect the IR sensor, Atmega328p Arduino board and ESP8266 module.	
Postcondition	Sending a signal to the Blynk application	
Primary actor	IR sensors, ATmega328p Arduino board, ESP8266 module.	
Trigger	Decide if the signal is coming from the Blynk application or not.	
Main Scenario	Step 1	Connect the IR sensors to the Arduino board.
	Step 2	Code the Arduino Atmega328p to detect cars using C programming language.
	Step 3	Set a particular point of sensitivity to detect cars.
	Step 4	Code the ESP8266 module to connect and display slots using the C programming language on an application.
	Step 5	Code the ESP8266 to send a signal to the Blynk application.
Extension	Step 2.1	Fail to detect cars.
	Step 5.1	Sending a signal to the Blynk application may fail.

Table 8 : Use Case Scenario 3

Number	03	
Name	Automatic Gate Opening System Implementation	
Summary	The method of open gates is by implementing IR sensors, an RFID reader, and an Atmega328p Arduino board.	
Precondition	Before the coding, connect the IR sensor, RFID reader and Atmega328p Arduino board.	
Postcondition	Sending a signal to the Servo Motors.	
Primary actor	Servo Motors, IR sensors, RFID reader, ATmega328p Arduino board.	
Trigger	Decide if the signal coming from Servo Motors or not.	
Main Scenario	Step 1	Connect the IR sensors and RFID reader to the Arduino board.
	Step 2	Code the Arduino Atmega328p to operate servo motors using C programming language.
	Step 3	Set a particular point of sensitivity to detect cars. (For Exit gate only)
	Step 4	Set a particular point of sensitivity and authority for RFID. (For Open gate only)
Extension	Step 3.1	Fail to detect cars.
	Step 4.1	Fail to detect authorize cars.

Table 9 : Use Case Scenario 4

Number	04	
Name	Automatic Night Lighting System Implementation	
Summary	The method of lighting up LEDs is by the implementation of the LDR sensor and Atmega328p Arduino board.	
Precondition	Before the coding, connect the LDR sensor and the Atmega328p Arduino board.	
Postcondition	Sending a signal to the LEDs.	
Primary actor	LEDs, LDR sensor, ATmega328p Arduino board.	
Trigger	Decide if the signal is coming from LEDs or not.	
Main Scenario	Step 1	Connect the LDR sensor and LEDs to the Arduino board.
	Step 2	Code the Arduino Atmega328p to detect dark or brightness using C programming language.
	Step 3	Set a particular point of sensitivity to detect dark or brightness.
	Step 4	Code the Arduino to send a signal to the LEDs.
Extension	Step 3.1	Fail to dark or brightness.
	Step 4.1	Sending a signal to the LEDs may fail.

Table 10 : Use Case Scenario 5

Number	05	
Name	RFID Authentication System Implementation	
Summary	The method of Authentication is by implementing an RFID reader and Atmega328p Arduino board.	
Precondition	Before the coding, connect the RFID reader and the Atmega328p Arduino board.	
Postcondition	Sending a signal to the Servo Motors.	
Primary actor	Servo Motors, RFID reader, ATmega328p Arduino board.	
Trigger	Decide if the signal is coming from an RFID reader or not.	
Main Scenario	Step 1	Connect the RFID reader to the Arduino board.
	Step 2	Code the Arduino Atmega328p to authenticate users using C programming language.
	Step 3	Set a particular point of sensitivity to detect and authenticate users.
	Step 4	Code the Arduino to send a signal to the Servo Motors.
Extension	Step 2.1	Fail to detect and authenticate users.
	Step 4.1	Sending a signal to Servo Motors may fail.

2.1.4 User case diagram

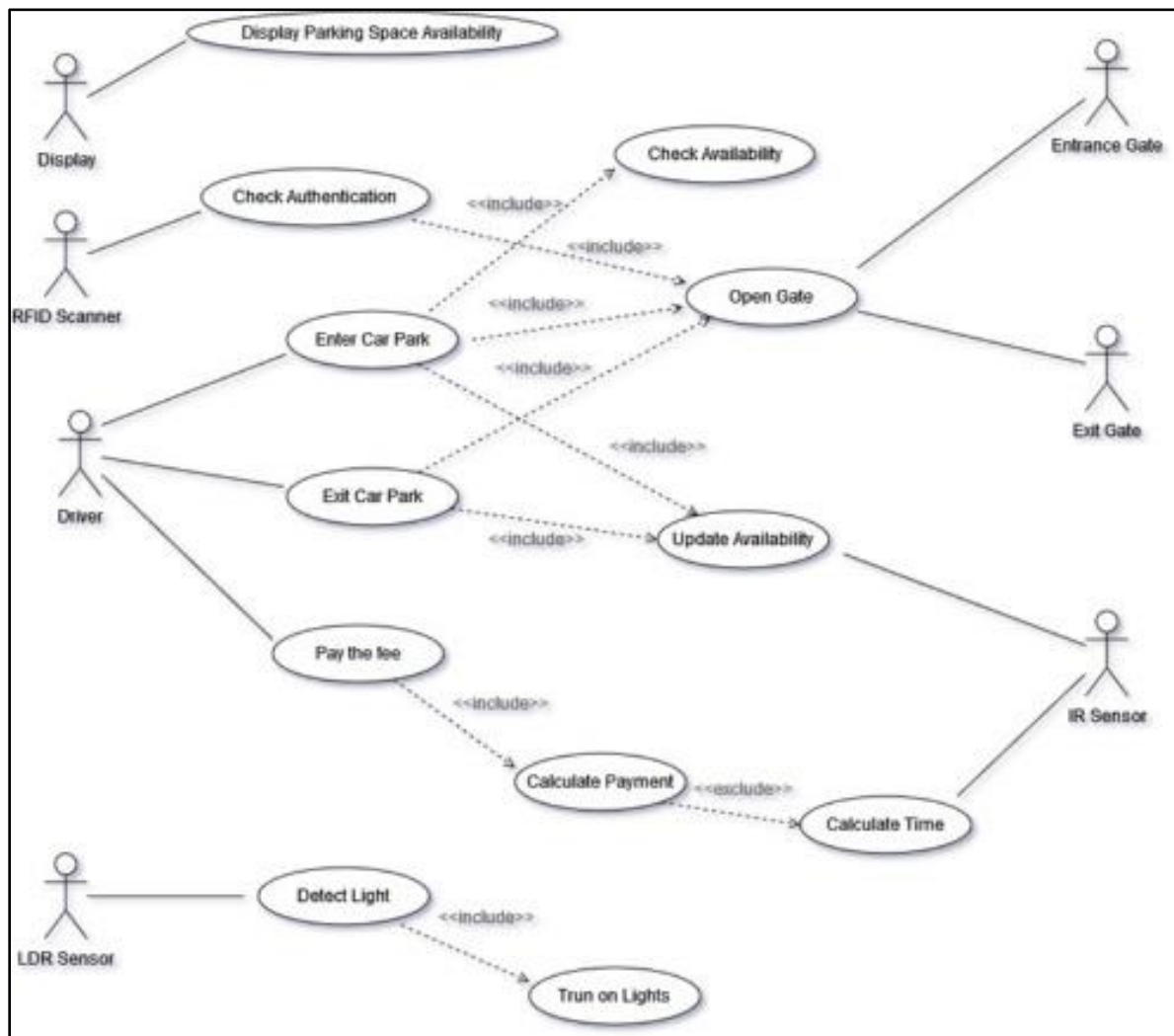


Figure 2.1 : User case diagram

2.2 Design

2.2.1 High Level Architecture Diagram

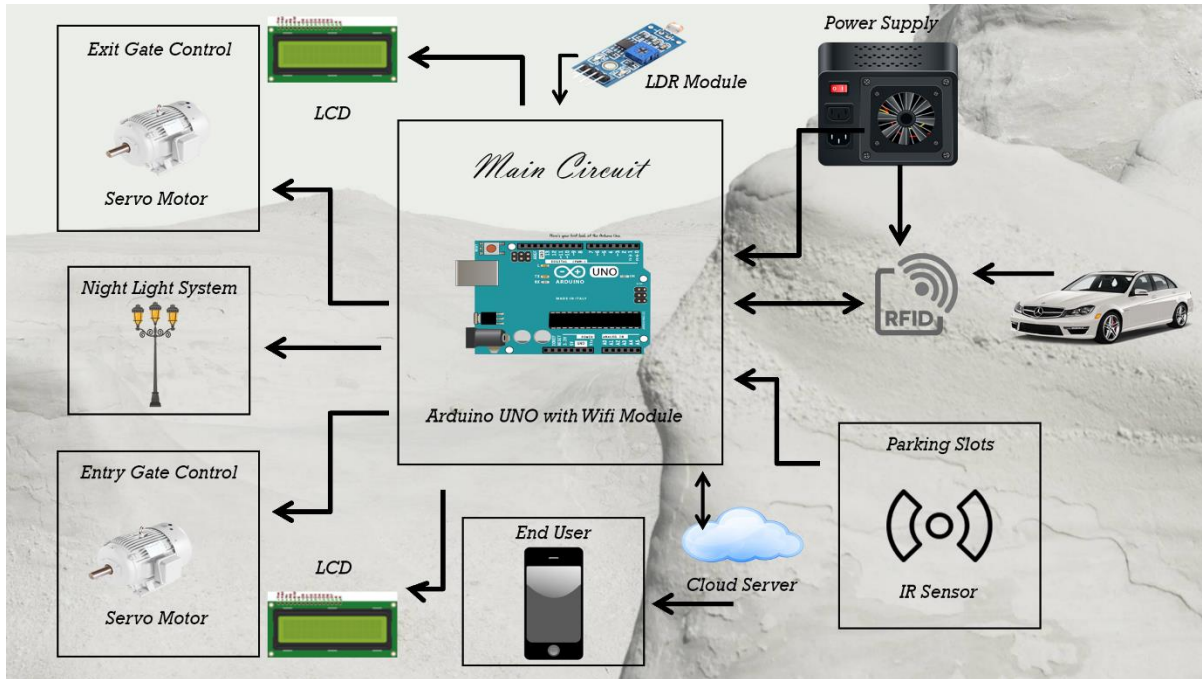


Figure 2.2 : High Level Architecture Diagram

The IoT intelligent car parking system's high-level architecture diagram comprises several essential elements. The Central System, the central hub for data processing and management, is at the system's center. It gets data from sensors placed in each parking space, which recognize the presence of cars and report the occupancy level to the central system. Users can monitor real-time parking availability, authenticate using RFID cards, make payments, and access navigation services using a User Interface, often in a mobile application or dedicated kiosks. This high-level architecture diagram depicts the data flow and interactions between the various components, enabling effective and user-friendly intelligent car parking.

2.2.2 Activity Diagram of Entire System

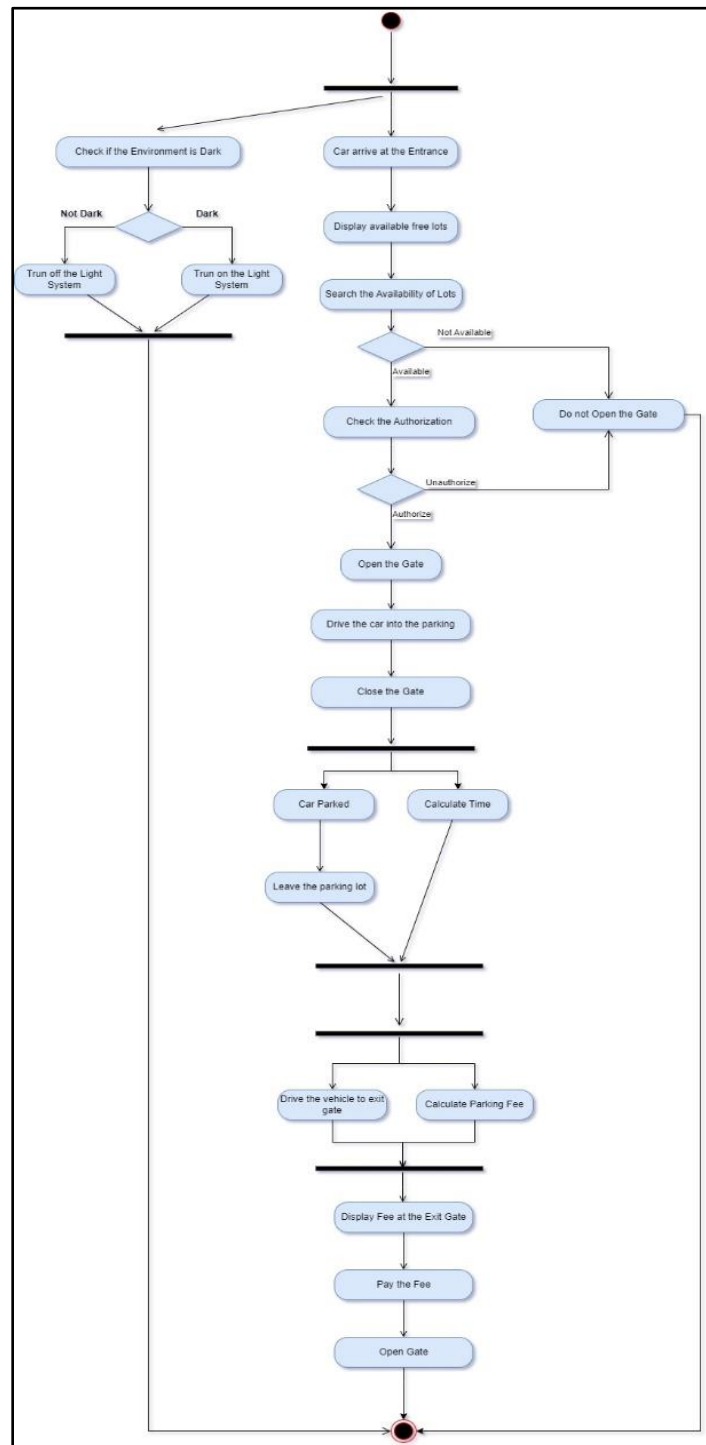


Figure 2.3 : Activity Diagram

2.2.3 Circuit Diagram

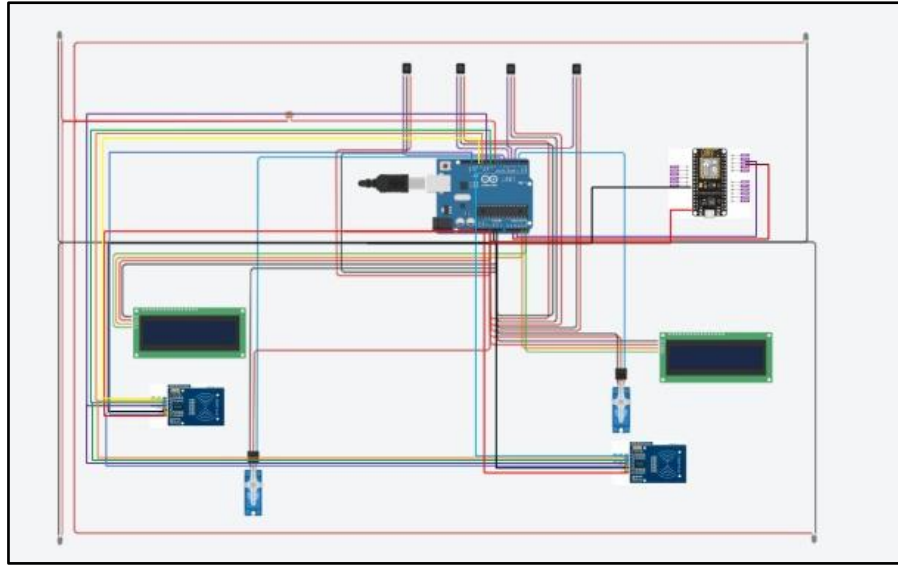


Figure 2.4 : Virtual Circuit Diagram

The system's virtual circuit diagram is visible in the image above. In this system, sensors and modules are the two main hardware categories. The sensors and modules used in this system are the ESP8266 Wi-Fi module, IR Sensors, RFID Scanner, and LDR Sensor. The Arduino Uno board is centrally connected to them. The circuit's Wi-Fi module transmits a signal that links the software and hardware interfaces.

2.2.4 User Interfaces

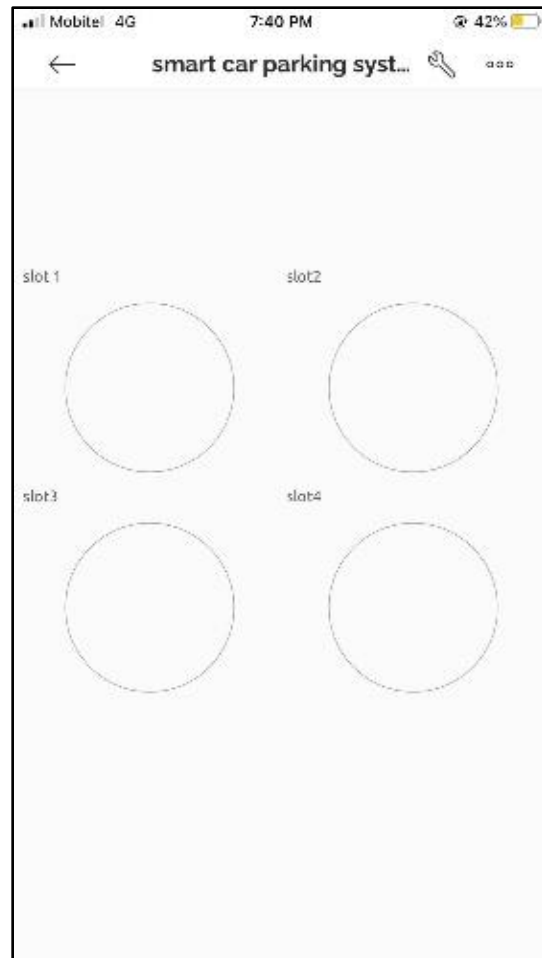


Figure 2.5 : User Interfaces

The sole user interface (UI) offered is for the Blynk app for smartphones' notifications feature. The parking lots are displayed in real-time. It turns red if there are no parking spaces available.

2.3 Implementation

2.3.1 Development tools

2.3.1.1 *Arduino IDE*

The Arduino IDE is one of the development tools utilized for this project. The Arduino Integrated Development Environment (IDE) is a software platform that serves as the primary programming environment for Arduino boards. It offers an intuitive user interface and a full complement of tools to make creating and uploading code to Arduino microcontrollers easier. Windows, Mac OS X, and Linux are just a few operating systems the IDE application is compatible with. The Arduino IDE supports a simplified version of the C++ programming language, making it accessible even to beginners in electronics and programming. Tools like code highlighting, syntax checking, and auto-completion provide a streamlined workflow that makes creating and editing code faster. Additionally, the Arduino IDE has a built-in serial monitor that shows real-time data and messages delivered from the Arduino board, enabling programmers to monitor and debug their code. Overall, the Arduino IDE makes it easier to construct Arduino projects, making it a valuable and adaptable tool for both professionals and enthusiasts.

2.3.1.2 *Blynk Cloud*

According to our preferences for the project, the Blynk cloud is used to construct the Blynk app. Blynk Cloud is a cloud-based platform that enables easy and secure connectivity between Internet of Things (IoT) devices and mobile applications. It offers a full range of tools and services that make building and implementing IoT projects easier. Developers may design interactive user interfaces and control panels for their IoT devices using the drag-and-drop interface of Blynk Cloud to create customized mobile apps. Blynk Cloud handles the communication between IoT devices and mobile apps, ensuring a seamless and secure connection through encrypted protocols. To build a graphical interface, this application compiles and provides the necessary addresses on the accessible widgets. To control NodeMCU in this project, we used Blynk Cloud. The system displays the open spaces in the parking lot.

2.3.2 Implementation Language

2.3.2.1 Arduino Language

The Arduino programming language is a streamlined version of C/C++ created especially for programming Arduino boards. It offers a user-friendly and approachable method for developing electronic devices. Since the language retains the fundamental grammar and organization of C/C++, programmers familiar with these languages can quickly move to Arduino programming.

2.4 Testing

Table 11 : Test scenario 1

Test Scenario ID		Slot Monitoring System Implementation - 1		Test Case ID	Slot Monitoring - 1	
Test Case Description		IR sensor sensitivity		Test Priority	High	
Pre-Requisite		Power on circuit		Post-Requisite	Connection to LCD	
Test Execution Steps:						
S.No	Action	Inputs	Expected Output	Actual Output	Test Result	
01.	Making an IR sensor sensitivity below the pre-defined sensitivity	Detection	No changes of the IR sensor, LCD	No changes of the IR sensor, LCD	Pass	
02.	Making an IR sensor sensitivity over the pre-defined sensitivity	Detection	Display the slots on LCD in real-time	Display the slots on LCD in real-time	Pass	
Test Scenario ID		Slot Monitoring System Implementation - 1		Test Case ID		Slot Monitoring - 2
Test Case Description		Connection to LCD		Test Priority		High
Pre-Requisite		Power on circuit		Post-Requisite		N/A
Test Execution Steps:						
S.No	Action	Inputs	Expected Output	Actual Output	Test Result	
01.	LCD not connected	Detection	Not Receiving a Result to LCD	Not Receiving a Result to LCD	Pass	

02.	LCD connected	Detection	Receiving a Result to LCD	Receiving a Result to LCD	Pass
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Table 12 : Test scenario 2

Test Scenario ID		Display slots on the Blynk application in real-time - 2		Test Case ID	Blynk application - 1	
Test Case Description		Display the slots		Test Priority	High	
Pre-Requisite		Connect to Wi-Fi		Post-Requisite	N/A	
Test Execution Steps:						
S.No	Action	Inputs	Expected Output	Actual Output	Test Result	
01.	Connect to Wi-Fi	IR Signal	Display the status of the slots in real-time on the Blynk application	Display the status of the slots in real-time on the Blynk application	Pass	
02.	Not Connect to Wi fi	IR Signal	Not display the status of the slots in real-time on the Blynk application	Not display the status of the slots in real-time on the Blynk application	Pass	

Test Scenario ID		Automatic Gate Opening System Implementation - 1		Test Case ID	Gate Opening - 1
Test Case Description		IR sensor sensitivity		Test Priority	High
Pre-Requisite		Power on circuit		Post-Requisite	Connection to Servo Motors
Test Execution Steps:					
S.No	Action	Inputs	Expected Output	Actual Output	Test Result
01.	Making an IR sensor sensitivity below the pre-defined sensitivity	Car Detection	Gate does not operate	Gate does not operate	Pass
02.	Making an IR sensor sensitivity over the pre-defined sensitivity	Car Detection	Gate should operate	Gate should operate	Pass
Test Scenario ID		Automatic Gate Opening System Implementation - 1		Test Case ID	Gate Opening - 2
Test Case Description		Connection to RFID reader		Test Priority	High
Pre-Requisite		Power on circuit		Post-Requisite	Connection to Servo Motors
Test Execution Steps:					
S.No	Action	Inputs	Expected Output	Actual Output	Test Result
01.	RFID card is authorized	RFID card	Gate should operate	Gate should operate	Pass

02.	RFID card is not authorized	RFID card	Gate does not operate	Gate does not operate	Pass
Test Scenario ID	Automatic Gate Opening System Implementation - 1		Test Case ID	Gate Opening - 3	
Test Case Description	Slot availability		Test Priority	High	
Pre-Requisite	Power on circuit		Post-Requisite	Connection to Servo Motors	
Test Execution Steps:					
S.No	Action	Inputs	Expected Output	Actual Output	Test Result
01.	Slots available	RFID card	Gate should operate	Gate should operate	Pass
02.	Slots not available	RFID card	Gate does not operate	Gate does not operate	Pass

Table 13 : Test scenario 3

Test Scenario ID		Automatic Night Lighting System Implementation - 4		Test Case ID	Night Lighting - 1
Test Case Description		LDR sensor sensitivity		Test Priority	High
Pre-Requisite		Power on circuit		Post-Requisite	Connection to LEDs
Test Execution Steps:					
S.No	Action	Inputs	Expected Output	Actual Output	Test Result
01.	Making an LDR sensor sensitivity below the pre-defined sensitivity	Detection	No changes of the LDR sensor LEDs	No changes of the LDR sensor LEDs	Pass
02.	Making an LDR sensor sensitivity over the pre-defined sensitivity	Detection	LEDs light up when it detects dark.	LEDs light up when it detects dark.	Pass

3. Conclusion

3.1 Limitations of developed objective

3.1.1 Integration with Existing Infrastructure

The integration of the already-existing parking infrastructure or systems is not a specific target. It is crucial to consider whether any pre-existing parking management systems, payment gateways, or security systems can seamlessly integrate with the IoT intelligent auto parking system. Failure to address this limitation could result in disruptions or inconsistencies in the parking management process.

3.1.2 Security Measures

While the objectives mention RFID authentication for vehicle entry and exit, there may be a need for comprehensive security measures outlined. Strong security measures should be incorporated into the system to guard against unauthorized access, data breaches, or tampering. Failure to solve these security issues could jeopardize the parking system's integrity and user data privacy and security.

3.1.3 User Adoption and Training

The objectives may only partially encompass factors related to user adoption and training. Successful implementation of the IoT intelligent car parking system relies on user acceptance and proper training to ensure efficient system utilization. To overcome potential difficulties or reluctance to change, it is crucial to consider user experience and usability and provide sufficient training and assistance to parking attendants and system administrators.

3.1.4 Environmental Factors

The developed objectives do not explicitly address environmental factors such as extreme weather conditions, power supply stability, or physical durability of the IoT infrastructure. These elements

may limit the system's performance and dependability. Thus, they should be considered during the project's planning and design phases.

3.2 Assessment of the project

The functionality, user experience, performance, security, impact on parking management, cost-effectiveness, and future concerns are all part of the IoT intelligent car parking system project assessment. The evaluation aims to ascertain whether the system properly conducts RFID authentication and fee calculation, reliably detects the availability of parking spaces, and fulfils its intended functionality. The system's performance in terms of response time, reliability, and scalability is evaluated, along with its security measures to ensure the protection of user data and compliance with privacy regulations. The project's cost-effectiveness is assessed by weighing possible cost savings and improved income against the advantages and value supplied. Based on feedback from stakeholders and cutting-edge technology, future considerations are identified to investigate potential areas for additional development and expansion. Overall, the assessment provides valuable insights into the project's success, areas for improvement, and potential for future enhancements.

3.3 Future works

3.3.1 Development of a Server

Only nearby mobile devices that are present can connect to and access the System. Therefore, setting up a server would be perfect for providing access to the System from a much greater distance for the user.

3.3.2 Reservation and Payment Integration

Integrate online payment and reservation features into the System to enable users to reserve parking spaces in advance. This feature can make users more comfortable, shorten wait times, and make payments more manageable.

3.3.3 Real-Time Navigation and Guidance

Enhance the user interface to provide real-time navigation and guidance to available parking spaces. The device can direct drivers to the closest open parking space by merging with GPS and mapping technology, minimizing search time and easing traffic congestion.

3.3.4 Mobile App Enhancements

Continuously improve the mobile app interface and user experience based on user feedback and emerging design trends. To increase customer engagement and happiness, provide features like personalized parking preferences, parking history, and notifications/alerts.

3.3.5 Smart Maintenance and Monitoring

Utilize remote monitoring and predictive maintenance tools to identify potential problems or malfunctions in the parking system and take appropriate action. This proactive approach minimizes downtime, optimizes system performance, and reduces maintenance costs.

3.4 Lessons learned.

3.4.1 Comprehensive Planning

Planning carefully and analyzing requirements are essential for a project's successful implementation. Setting reasonable expectations and preventing scope creep is made more accessible by clearly outlining the project's objectives, scope, and deliverables.

3.4.2 Effective Communication

Maintaining clear and consistent communication among team members, stakeholders, and end-users fosters collaboration, minimizes misunderstandings, and ensures alignment with project objectives.

3.4.3 Rigorous Testing and Quality Assurance

A thorough testing and quality assurance methodology must be put in place to find and fix any system flaws, defects, or performance problems. Thorough testing at each stage helps ensure a reliable and robust system.

3.4.4 Continuous Improvement

The IoT intelligent car parking system may be continuously improved by incorporating input and adopting a culture of continuous improvement. Regularly seeking user feedback and staying updated with industry trends ensures the plan remains relevant and valuable.

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Appendix A: Design Diagrams

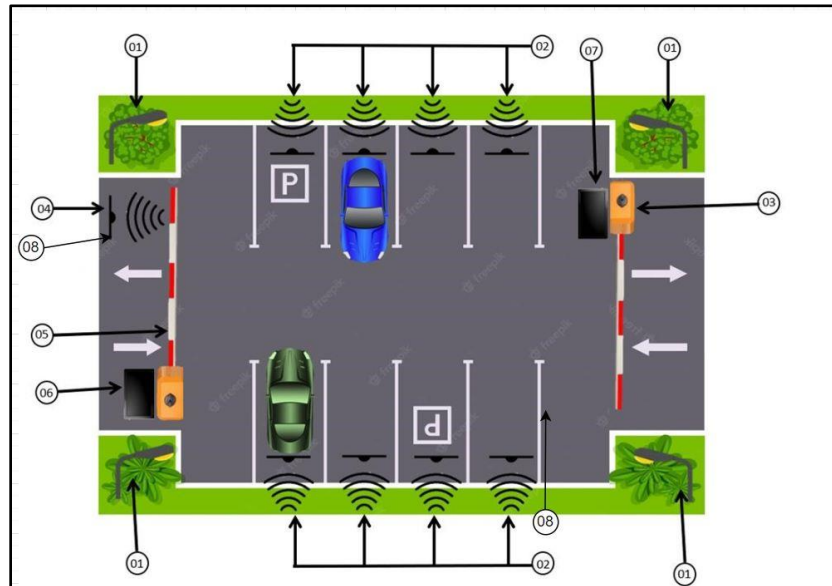


Figure 4.1 : Design Diagrams

01 – Automatic Lights

02 – IR Sensors

03 – Exit Gate

04 – IR Sensor at the Entrance

05 – Entrance Gate

06 – Displaying available Slots

07 - Parking Fee Display

08 - RFID Card Scanner

Appendix B: Physical Design



Figure 4.2 : Physical Design